Statistical Estimation for BGG Ratings

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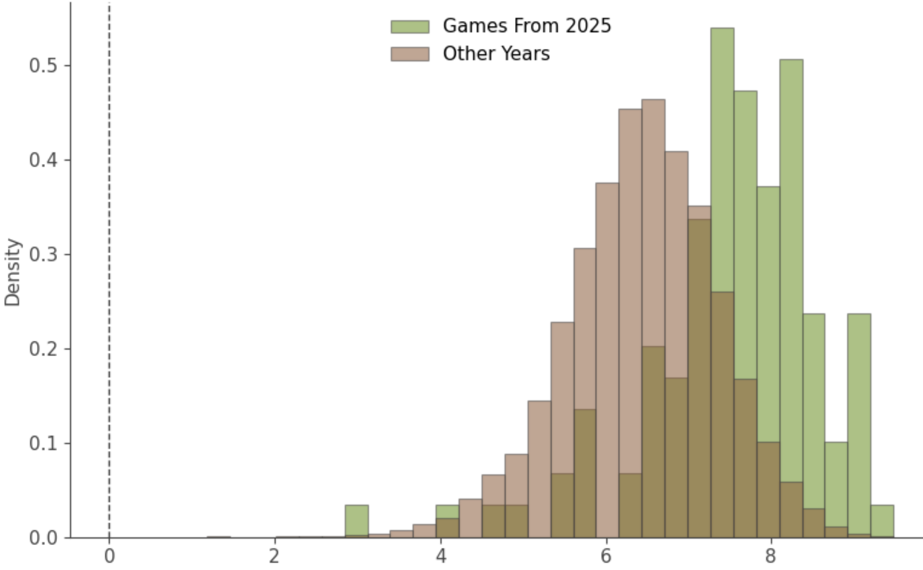
Data

- Collected in August 2025 and March 2026
- Source: BoardGameGeek XML API
- Processing code: GitHub Repository
- A much more minimal copy can be downloaded by pressing this link

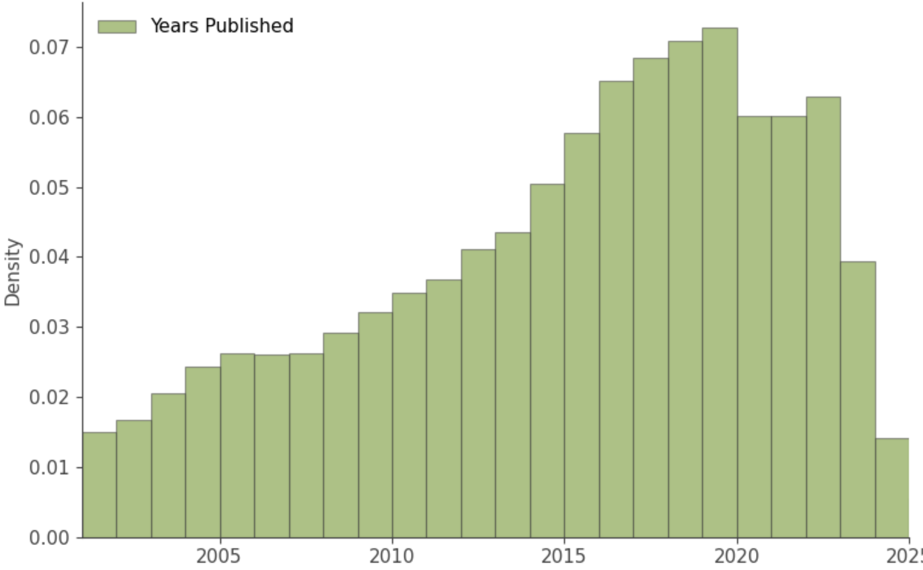
Fetching Game Data from BoardGameGeek

```
def fetch_things(
    ids: Iterable[int], max_retries: int, retries: int
) -> list[dict]:
    url = f"‘url‘/thing?id={’,’.join(map(str, ids))}&stats=1"
    retries = 0
    while retries < max_retries:
        response = requests.get(url)
        if response.status_code == 200:
            return parse(response.text)["items"]["item"]
        else:
            time.sleep(5)
    raise Exception()
```

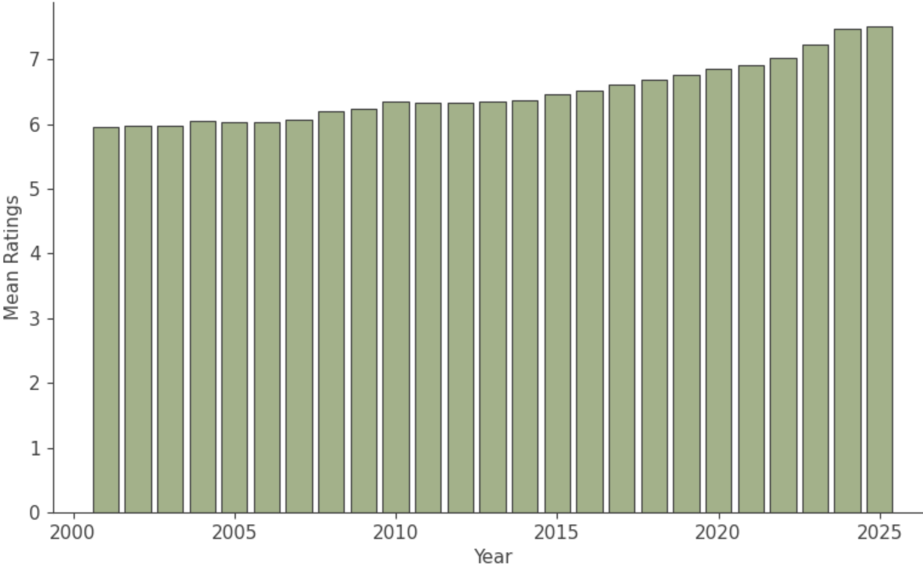
Distribution of Ratings



Games Published Per Year



Mean Rating by Year



Confidence Interval for Mean Rating

Goal: Estimate a confidence interval for the true mean rating of a game.

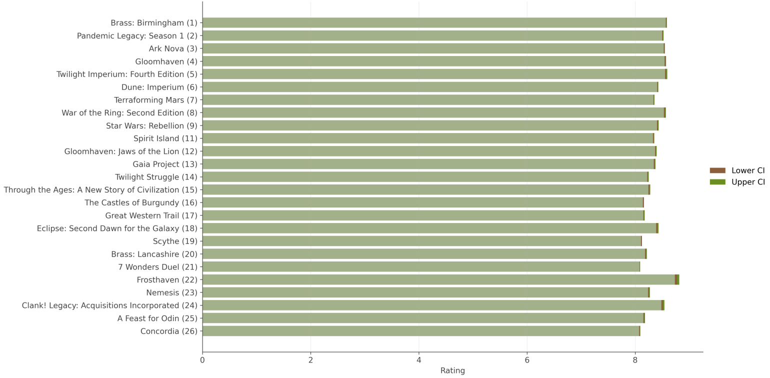
- Uses the **t-distribution** because the population standard deviation is unknown.
- Standard error is computed from the sample ratings.
- Degrees of freedom: $df = n - 1$

Implementation

`se = self.`$_{s}tandard_{e}rror()df = self.usersrated - 1t_{c}rit = stats.t.ppf((1 + confidence)/2, df)$

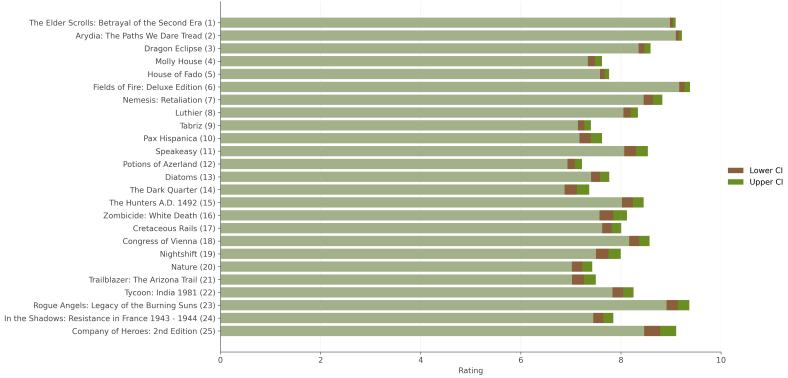
`margin = t`$_{c}rit * sereturn(self.average - margin, self.average + margin)$

Top 25 Games



Brass: Birmingham (1)
Pandemic Legacy: Season 1 (2)
Ark Nova (3)
Gloomhaven (4)
Twilight Imperium: Fourth Edition (5)
Dune: Imperium (6)
Terraforming Mars (7)
War of the Ring: Second Edition (8)
Star Wars: Rebellion (9)
Spirit Island (11)
Gloomhaven: Jaws of the Lion (12)
Gaia Project (13)
Twilight Struggle (14)
Through the Ages: A New Story of Civilization (15)
The Castles of Burgundy (16)
Great Western Trail (17)
Eclipse: Second Dawn for the Galaxy (18)
Scythe (19)
Brass: Lancashire (20)
7 Wonders Duel (21)
Frosthaven (22)
Nemesis (23)
Clank! Legacy: Acquisitions Incorporated (24)
A Feast for Odin (25)
Concordia (26)
0
2
4
6
8
Rating
Lower CI
Upper CI

Top 25 Games vs. 2025 Releases



Updated 2025 Comparison